NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*COSC(MODEL-A&APEX)	RTA-1	Date: 08-02-26
Time: 3 Hrs	2025_P1	Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**B**	2	**D**	3	**B**	4	**D**	5	**BCD**
6	**AB**	7	**BCD**	8	**1**	9	**40**	10	**1**
11	**258**	12	**3**	13	**1**	14	**B**	15	**B**
16	**C**								

PHYSICS

17	**C**	18	**B**	19	**B**	20	**B**	21	**ABCD**
22	**ABD**	23	**AD**	24	**14.80**	25	**10**	26	**8**
27	**4.70**	28	**2.20**	29	**1.10**	30	**C**	31	**C**
32	**C**								

CHEMISTRY

33	**B**	34	**C**	35	**A**	36	**A**	37	**ACD**
38	**ABC**	39	**BCD**	40	**942**	41	**8**	42	**3**
43	**17**	44	**7**	45	**25**	46	**D**	47	**B**
48									

For More Join: @IITJEE_Advanced

SOLUTIONS

MATHS

1. We know, $\sin 3\theta = \sin\theta(1+2\cos 2\theta)$

$$\Rightarrow \sin x = \sin\left(\frac{x}{3}\right).\left(1+2\cos\frac{2x}{3}\right) \Rightarrow \frac{\sin x}{x} = \prod_{k=1}^{\infty}\left(\frac{1+2\cos\left(\frac{2x}{3^k}\right)}{3}\right) = f(x)$$

Therefore the given expression is not differentiable at $\{\pi, 2\pi, 1, 2, 3\}$. The number of points are 5

2. $y^{11}(0)$, for $\left(e^{\sin x}-1\right)^4 + (\cos x - 1)^3 e^{\sin x} + \left(e^x - e^{-x}\right)^3$ is always gives zero at $x = 0$

3. $$\frac{\pi}{2} - \cos^{-1}\cos\left[\frac{2\left(x^2+5|x|+3\right)-2}{\underbrace{x^2+5|x|+3}_{0<\downarrow_2<2}}\right] = \cot\cot^{-1}\left(\frac{2}{9|x|}-2\right)+\frac{\pi}{2}$$

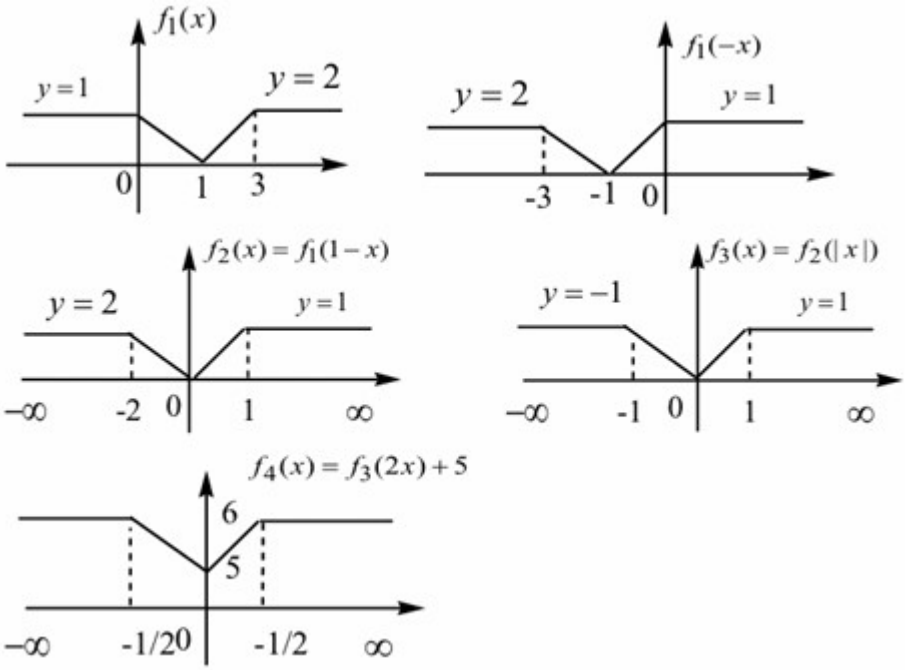
$$\frac{\pi}{2} - 2 + \frac{2}{x^2+5|x|+3} = \frac{2}{9|x|} - 2 + \frac{\pi}{2} \qquad \Rightarrow |x|^2 - 4|x| + 3 = 0$$

$$|x| = 1, 3 \qquad \Rightarrow x = \pm 1, \pm 3$$

4. Use the function $f(x) = \left(\cot^{-1}(x)\right)^2 + \frac{2}{\sqrt{x^2+1}}$

5. $$f_1(x) = \begin{cases} 1 & , x \in (-\infty, 0) \\ |x-1| & , x \in [0,3] \\ 2 & , x \in (3,8) \end{cases}$$

$$f_2(x) = f_1(1-x) = f_1(-(x-1))$$



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6. As maximum value of $\frac{x^2-2x+d}{x^2+3x+d}$ is 6

$\Rightarrow \frac{x^2-2x+d}{x^2+3x+d}=6$ has equal roots

$\Rightarrow 5x^2+20x+5d=0 \Rightarrow 16-4d=0 \Rightarrow d=4$ and $a=\frac{2}{4}$

7. Apply $R_3 \to R_3-R_2; R_2 \to R_2-R_1 \Rightarrow f(x)=-2d^2$

(Where d is C.D. of AP)

$\therefore d^2=1 \Rightarrow =\pm 1 \Rightarrow d_1=-1$ & $d_2=1$ & $g(x)=x^2-1$

8. $f(x)=\sin^{-1}(x+2019)-\cos^{-1}(x+2021)$

$-1\le x+2019\le 1$ $\quad -1\le x+2021\le 1$

$-2020\le x\le -2018$ $\quad -2022\le x\le -2020$

$\therefore$ Domain of $f(x)$ is $\{-2020\}$

$g(x)=\sin^{-1}(x+2017)+\cos^{-1}(x+2019)$

$-1\le x+2017\le 1$ $\quad -1\le x+2019\le 1$

$-2018\le x\le -2016$ $\quad -2020\le x\le -2018$

$\therefore$ Domain of $f(x)$ is $\{-2018\}$

$x_1=-2020$ and $x_2=-2018$

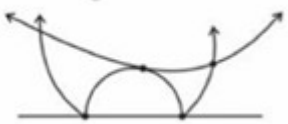
$\lim\limits_{x\to -4038}\frac{\sin(x+4038)-\sin(2x+8076)}{\tan(x+4038)}=1$

9. Case – I: When for 3 solution $y=x^2-4x+a$ touches

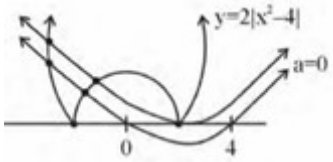
$y=8-2x^2$ in $[-2,2] \Rightarrow x^2-4x+a=8-2x^2$

$\Rightarrow 3x^2-4x+a-8=0$ has exactly one root $16-12(a-8)=0$

$\Rightarrow 4-3a+24=0 \Rightarrow 28=3a \quad \Rightarrow a=\frac{28}{3}$



(note: L.R. of $y=x^2-4x+a>$ L.R. of $y=2x^2-8$)



When $y=x^2-4x+a$ passes through $(2,0)a=4$

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10. Let point of intersection of curves is (x_1, y_1) and let $2\cot\sqrt{\frac{\pi^2}{16}-a^2}=k$

Now differentiate $\log\left(x^2+y^2\right)=k\tan^{-1}\frac{y}{x}$ w.r.t x w get

$$\frac{1}{x^2+y^2}\left(2x+2y\frac{dy}{dx}\right)=k\left(\frac{1}{1+\frac{y^2}{x^2}}\right)\left\{\frac{x\frac{dy}{dx}-y}{x^2}\right\}$$

$$\Rightarrow\left\{x+y\frac{dy}{dx}\right\}=k\left\{x\frac{dy}{dx}-y\right\}\Rightarrow\frac{dy}{dx}=\frac{2x+ky}{kx-2y}$$

$$\Rightarrow\left.\frac{dy}{dx}\right]_{(x_1,y_1)}=\frac{2x_1+ky_1}{kx_1-2y_1}=m_1 \qquad \Rightarrow\frac{dy}{dx}\text{ for } y=mx \text{ is } m=\frac{y_1}{x_1}$$

$$\Rightarrow\tan\theta=\left|\frac{\frac{2x_1+ky_1}{kx_1-2y_1}-\frac{y_1}{x_1}}{1+\left(\frac{2x_1+ky_1}{kx_1-2y_1}\right)\left(\frac{y_1}{x_1}\right)}\right|=\left|\frac{2x_1^2+kx_1y_1-kx_1y_1+2y_1^2}{kx_1^2-2x_1y_1+2x_1y_1+ky_1^2}\right|=\left|\frac{2}{k}\right|$$

$$\Rightarrow\tan\theta=\tan\sqrt{\frac{\pi^2}{16}-\alpha^2}\Rightarrow\theta=\sqrt{\frac{\pi}{16}-\alpha^2}$$

$\Rightarrow$ max value of θ is $\frac{\pi}{4}$

11. $g\left(x^3-1\right)=x^5+x-2 \qquad \Rightarrow g(t)=(t+1)^{\frac{5}{3}}+(t+1)^{\frac{1}{3}}-2$

$$g(t)=\frac{5}{3}(t+1)^{\frac{2}{3}}+\frac{1}{3}(t+1)^{\frac{-2}{3}} \qquad g(7)=32, g'(7)=\frac{27}{4}$$

$$\frac{d}{dx}\left(\left(x^3+1\right)g\left(x^3\right)\right)=\left(x^3+1\right)g'\left(x^3\right)3x^2+g\left(x^3\right).3x^2$$

$$\left.\frac{d}{dx}\left(\left(x^3+1\right)g\left(x^3\right)\right)\right|_{x=7^{\frac{1}{3}}}=l=3.7^{2/3}.86\Rightarrow l\left(7^{2/3}\right)=258$$

12. $$f(x)=\begin{cases}\frac{1}{x}; & if\ x^2>1 \quad \Rightarrow x<-1\ or\ x>1\\ ax^3+bx^2; & if\ 0\le x^2<1\Rightarrow -1<x<1\\ \frac{1/x+ax^3+bx^2}{2}; & if\ x^2=1\end{cases}$$

$\because$ f is continuous $\therefore$ at $x=1$ and at $x=-1$

$1=a+b, -1=-a+b$ and $a=1$

$\therefore$ point A and B are $=(-1,3)$ and $(1,-1)$

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$\therefore\ g(x)=\lambda(x-1)(x+1)$ $\qquad g(x)=\lambda\left(\frac{x^3}{3}-x\right)+c$

$g(-1)=\frac{2\lambda}{3}+c=3$ $\qquad g(1)=-\frac{2\lambda}{3}+c=-1$

$c=1$ and $\lambda=3$ $\qquad \therefore g(x)=x^3-3x+1 \quad \therefore g(2)=3$

13. $f(xy)=f(x)+f(y)+\frac{1}{x}+\frac{1}{y}-\frac{1}{xy}\forall x,y>0 \quad \ldots(1)$

Differentiating w.r.t. y keeping x constant, we get

$xf'(xy)=f'(y)-\frac{1}{y^2}+\frac{1}{xy^2}$

Putting $y=1$, we get

$xf'(x)=f'(1)-1+\frac{1}{x}$

$\therefore xf'(x)=1+\frac{1}{x}$

$\Rightarrow f'(x)=\frac{1}{x}+\frac{1}{x^2}$

$\Rightarrow f(x)=\log_e x-\frac{1}{x}+C$

In (1), putting $x=y=1$, we get $f(1)=-1$

$\therefore C=0$

So, $f(x)=\log_e x-\frac{1}{x}$

14. (II) $f(x)=\cos^{-1}\left(\frac{x^2}{1+x^2}\right)=\cos^{-1}\left(1-\frac{1}{1+x^2}\right)$

$t=1-\frac{1}{1+x^2}\Rightarrow t\in[0,1)$ $\qquad \therefore \cos^{-1}t\in\left(0,\frac{\pi}{2}\right]=(a,b]\Rightarrow[a+b]=1\sqrt{a^2+b^2}$

(III) $12\left(\tan^2\left(\tan^{-1}\alpha\right)+\tan^2\left(\tan^{-1}\beta\right)\right)=12\left(\alpha^2+\beta^2\right)$

$=12\left((\alpha+\beta)^2-2\alpha\beta\right)$ $\qquad =12\left(\frac{9}{4}-2.1\right)=3$

(IV) $\sin\left(\cos^{-1}x\right)=\sin\left(\frac{\pi}{2}-\sin^{-1}x\right)=\cos\left(\sin^{-1}x\right)=t$

$f(x)=\sin\left(\cos^{-1}t+\sin^{-1}t\right)=\sin\frac{\pi}{2}=1$

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15. (P) $f(z)=\frac{16\times16}{12\times16}=8,\ f(4)=\frac{8\times8}{2\times8}=4$

(Q) $f(x)=2x+2$

(R) $f(x)=x^2+xe^x$

(S) $\left(f(y)-f(x)\right)y^y=x^x f\left(\frac{y^y}{x^x}\right)$ differentiate w.r.t x keeping y constant

$$-f'(x).y^y=x^x f\left(\frac{y^y}{x^x}\right)(1+\ln x)-x^x f'\left(\frac{y^y}{x^x}\right)\frac{y^y(1+\ln x)}{x^x}$$

$f'(1)=1$, put $y^y=x^x$, we get $-f(x)\,x^x=x^x f(1)(1+\ln x)-x^x f'(1)(1+\ln x)$

$f(1)=0$, hence $f'(x)=1+\ln x\Rightarrow f(x)=x\ln x$

$\Rightarrow$ area bounded between $y=x\ln x$ and $y=2x-2x^2$ is $\frac{7}{12}$

16. (P) Continuity must b checked at $x=1,e,3,\sqrt{10},\sqrt{11},\sqrt{12},3.5$ Further, $f(1)=0$ and $\lim\limits_{x\to1}f(x)=\lim\limits_{x},\operatorname{sgn}(x-2)\times[\log x]=0$

Hence, $f(x)$ is continuous at $x=1$

$f(e)=1,\ f(e^-)=0$

Hence $f(x)$ is discontinuous at $x=e$

$\lim\limits_{x\to3^+}f(x)=\lim\limits_{x\to3^-}\operatorname{sgn}(x-2)\times[\log_e x]=1$ $\Rightarrow\lim\limits_{x\to3^+}f(x)=\lim\limits_{x\to3^+}\{x^2\}=0$

Hence, $f(x)$ is discontinuous at $x=3$

Also $\{x^2\}$ is discontinuous $x=\sqrt{10},\sqrt{11},\sqrt{12}$ therefore,

$\lim\limits_{x\to\frac{7^-}{2}}f(x)=\lim\limits_{x\to3^-}\{x^2\}=0.25=f(3.5)$

Hence, $f(x)$ is discontinuous at

$x=3,\sqrt{10},\sqrt{11},\sqrt{12},e$

(Q) clearly there is one point of inflection

(R) not differentiable at $x=1$ and $x=\sqrt{3}$

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PHYSICS

17. Since water in the smaller vessel is taking heat from the water in the large vessel. It can reach to a temperature 100^0C. Since the temperature of water in the large vessel cannot go beyond 100^0C before boiling completely. Hence water in the smaller vessel will not take heat from the water outside, it will not boil.

18. For $1\to 2$

$$\frac{\rho}{P}=\frac{\rho_0}{P_0}=\text{ Constant}$$

$$\frac{\rho M}{\rho RT}=\frac{\rho_0}{P_0}$$

i.e., $T=$ const. for $1\to 2$

$$W_{1\to 2}=\frac{nMP_0}{\rho_0}\ell n\left[\frac{P_0}{3P_0}\right]$$

$$W_{2\to 3}=P\Delta=nR\Delta T=nRV\frac{[3P_0]}{R}\left[\frac{1}{\rho_0}-\frac{1}{3\rho_0}\right]$$

$$=3P_0Mn\left[\frac{2}{3\rho_0}\right]=\frac{2P_0Mn}{\rho_0}$$

$$W_{3\to 1}=0$$

$$W_{cycle}=\frac{-P_0M}{\rho_0}\ell n3+\frac{2P_0M}{\rho_0}$$

19. $n=2_1C_v=\frac{3}{2}R$

$$\Delta Q=nC_v\Delta T$$

$P_1=P_0;\ P_2=P_0+kx/A$ & Volume is constant

$$\Rightarrow\frac{P_2}{P_1}=\frac{T_2}{T_1}=\frac{(P_0+kx/A)}{P_0}=\frac{\left(10^5+\frac{10^3}{10^{-2}}\right)}{10^5}=\frac{T_2}{300}\Rightarrow T_2=bcvk$$

$$\Delta Q=nC_v\Delta T=2\times\frac{3}{2}R300=900R$$

21. $P=K\frac{E}{V}$

$\therefore PV=KE$

Also for ideal gas $PV=nRT$

22. The thermal resistance of cylinder is

$$R=fdR=\int\frac{dr}{2kTr\ell}=\frac{1}{2Tk\ell}\ell n2$$

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Rate of heat flow $P=\frac{\Delta T}{R}=\frac{402Tk\ell}{\ell n2}$

$=\frac{80\pi\times0.693\times1}{0.693}=80\pi\ J/\sec$

Rate of ice melting $=L\frac{dm}{dt}=80\pi$

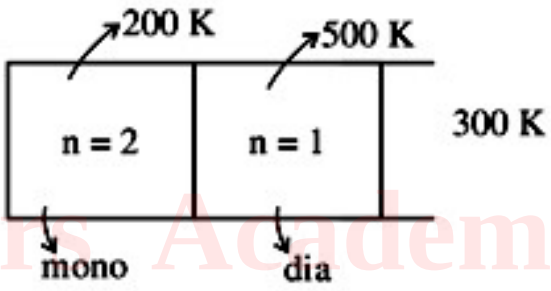
$\frac{dm}{dt}=\frac{80\pi}{804.2\times10^3}=\frac{\pi}{4200}kg/s$

D) $m=\pi\times10^{-4}\times900\times1\frac{\pi}{42w}\times t=\pi9\times10^{-2}\Rightarrow t=42.9$

23. $dQ_1=nC_p\Delta T=2\times\frac{5R}{2}\times[300-200]=5R\times100$

For compartment 2 $dQ=nC_p\Delta T$

$=-1\times\frac{7R}{2}\times200=-700R$



Work done by gas in compartment 1 $=P\Delta V=nR\Delta T=2R100=200R$

Heat lost by compartment 2 to its surrounding $=700R$

Work done by gas in compartment 2 on gas in compartment $1=-200\ R$

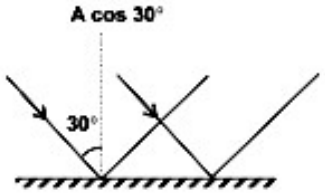
24. $\frac{dN}{dt}=r-\lambda N$

$\int_{N_0}^{N}\frac{dN}{r-\lambda N}=\int_0^t dt$

$\frac{dN}{dt}=r-\lambda N=(r-\lambda N_0)e^{-\lambda t}=\left(10^4-\frac{10^5}{20}\right)e^{-t/20}=\frac{5000}{7.4}=\left(\frac{10^4}{14.8}\right)$

26. I = rate of flow of energy through unit area $=nhv$, where n is the no. of photons following through unit area per sec. Rate of flow of momentum (momentum flux) through unit area $=\frac{nh}{\lambda}=\frac{nhv}{C}=\frac{I}{C}$.

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Consider a cylindrical beam of parallel rays whose section the mirror is A.

Rate of flow of momentum through this beam $= \frac{I}{C} A\cos\theta$

Normal component of the momentum flux $= \left(\frac{IA\cos\theta}{C}\right)\cos\theta$

Momentum flux of the reflected photon $= -.06\left(\frac{IA\cos^2\theta}{C}\right)$

Force = Rate of change of momentum $= (1.6)\frac{IA\cos^2\theta}{C}$

Pressure $= \frac{F}{A} = \frac{(1.6)I\cos^2\theta}{C} = \frac{1.6\times 2\times 10^3\times\frac{3}{4}}{3\times 10^8} = 8\times 10^{-6}\ N/m^2$

27. $\frac{13.6Z^2}{n^2} = 36\%$ of 13.6 … (i)

Also, ${}^nC_2 = 10$ … (ii)

and, $eV_0 = \frac{hc}{\lambda} - \phi$ … (iii)

28. $E = \sigma T^4$

$\Rightarrow \frac{E_2}{E_1} = \frac{T_2^4}{T_1^2} = 256 = 4^4$

$\Rightarrow T_2 = 4T_1$

Also, $\lambda_1 T_1 = \lambda_2 T_2$ gives $\lambda_2 = \frac{\lambda_1}{4} = 1000\ \overset{0}{A}$

$KE_{max} = \frac{hC}{\lambda_2} - \phi \Rightarrow \frac{12400}{1000} - \phi = 12.4 - \phi$ … (i)

$\Delta E_{2\to 4} = 13.6\times 2^2\times\left[\frac{1}{2^2} - \frac{1}{4^2}\right] = 13.6\times 4\times\frac{3}{16} = 10.2\,eV$ … (ii)

From (i) and (ii) $12.4 - \phi = 10.2$

$\Rightarrow \phi = 2.2\ eV$

29. ${}_1^2H + {}_1^2H \to {}_1^3H + {}_1^1P$

Using momentum conservation

$\vec{P}_1 + \vec{P}_2 = 0$

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$\vec{P}_1 = \vec{P}_2$

$P_1^2 = P_2^2$

$2m_1k_1 = 2m_2k_2$

$3k_1 = k_2$ .. (i)

Also,

$k_1 + k_2 = 8.4 - 4$

$k_1 + k_2 = 4.4$

Solving (i) and (ii) we get

$4k_1 = 4.4$

$k_1 = 1.1\ MeV$ and $k_2 = 3.3\ MeV$

Kinetic energy of the tritium, $k_1 = 1.1\ MeV$

Kinetic energy of the proton, $k_2 = 3.3\ MeV$

30. For He^+ absorption occur from level $(2 \to 4)$

For Li^{2+} absorption occur from level $(3 \to 6)$

31. (I) $Q = \int Tds =$ Area between T-s diagram $= 700\ J$

(II) $\Delta Q = n\frac{3}{2}R(T_f - T_i) + n\frac{5}{2}R(T_f - T_i)$

$= \frac{3}{2}(nRT_f - nRT_i) + \frac{5}{2}(nRT_f - nRT_i)$

$= \frac{3}{2}(500 - 100) + \frac{5}{2}(1000 - 500) = 1850\,J$

(III) Process is $T = 100V \Rightarrow \frac{T}{V} =$ constant i.e. pressure = constant

$\therefore Q = nC_P\Delta T = 2\frac{5}{2}R(300 - 100) = 1000\,R$

(IV) $A \to B$ is V = constant and $B \to C$, $P =$ constant

$\Delta Q = n\frac{3}{2}R(T_B - T_A) + n\frac{5}{2}R(T_C - T_B) = (1000 - 500) + \frac{5}{3}(500 - 1000)$

CHEMISTRY

33. Conceptual

34. $A = CaO;\ B = CaC_2;\ D = CaNCN;\ E = NH_3$

$Mg_3N_2 + 6H_2O \to 3Mg(OH)_2 + 2NH_3$

35. Conceptual

36. Number of moles of moles of H_2SO_4 present in the battery initially = 8 moles

Charge drawn from the battery $= \frac{1.93 \times 20 \times 60 \times 60}{96500} = 1.44\,F$

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$Pb + PbO_2 + 2H_2SO_4 \rightarrow 2PbSO_4 + 2H_2O$

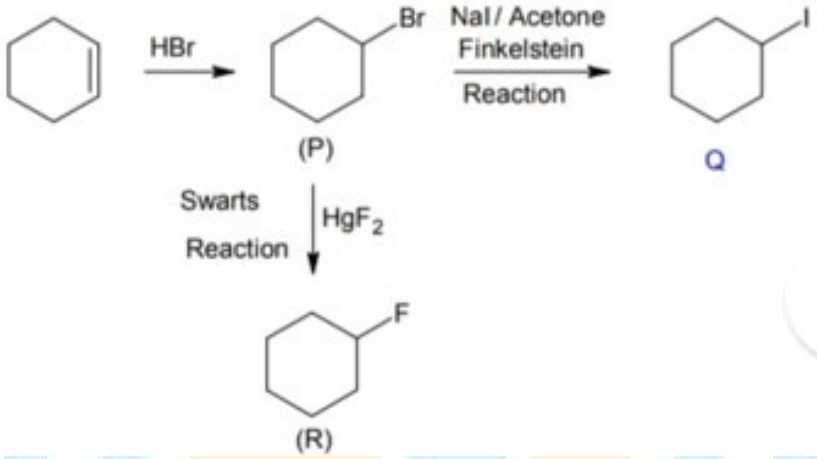
For consumption of 2 moles of H_2SO_4, the charge drawn is 2 Faradays

Number of moles of H_2SO_4 consumed $=1.44$

Number of moles of H_2SO_4 left $= 8 - 1.44 = 6.56$

Concentration of H_2SO_4 left in the battery $= \frac{6.56}{2} = 3.28M$

37.



C-X bond enthalpy in P, Q, R follows the order are R > P > Q

38. $6PCl_5 + P_4O_{10} \rightarrow 10POCl_3$

$PCl_5 + H_2O \rightarrow POCl_3 + 2HCl$

$PCl_5 + SO_2 \rightarrow POCl_3 + SOCl_2$

$PCl_3 + 3C_2H_5OH \rightarrow H_3PO_3 + 3C_2H_5Cl$

39. Check the E^0_{Cell} values. If it is positive, the cell reaction is spontaneous.

40. $P = CaC_2,\ Q = C_2H_2,\ R = C_6H_6,\ S = C_6H_5Br$

41. $X = 4, Y = 4$

42. Conceptual

43. $3.72 = 2 \times \frac{6 \times 1000}{M.Wt \times 200} \Rightarrow M.Wt. = 30$

$1.6 = \frac{30 \times 4}{a^3 \times 6 \times 10^{23}} \Rightarrow a^3 = 12.5 \times 10^{-23} = 125 \times 10^{-24} \Rightarrow a = 5 \times 10^{-8}\ cm = 500\ pm$

$a = 2(r_C + r_a) \Rightarrow 500 = 2(r_C + 170) \Rightarrow r_C = 80\ pm$

$X = \left(\frac{r_a}{r_C}\right) \times 8 = \left(\frac{170}{80}\right) \times 8 = 17$

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44. $R=\rho\times\frac{l}{A}\Rightarrow\frac{1}{R}=\frac{1}{\rho}\times\frac{1}{l/A}\Rightarrow\frac{1}{R}=\kappa\times\frac{A}{l}\Rightarrow\frac{1}{50}=\kappa\times\frac{10}{4}\Rightarrow\kappa=8\times10^{-3}\,Scm^{-1}$

$\lambda_m=8\times10^{-3}\times\frac{1000}{0.1}=80\,Scm^2mol^{-1}$

$\lambda_m^0=\frac{230}{2}+425-\frac{280}{2}=400\,Scm^2mol^{-1}$

$\alpha=\frac{\lambda_m}{\lambda_m^0}=\frac{80}{400}=0.2$

$Ka=\frac{C\alpha^2}{(1-\alpha)}=\frac{0.1\times(0.2)^2}{(1-0.04)}=\frac{0.1\times(0.04)}{0.96}=4.16\times10^{-3}$

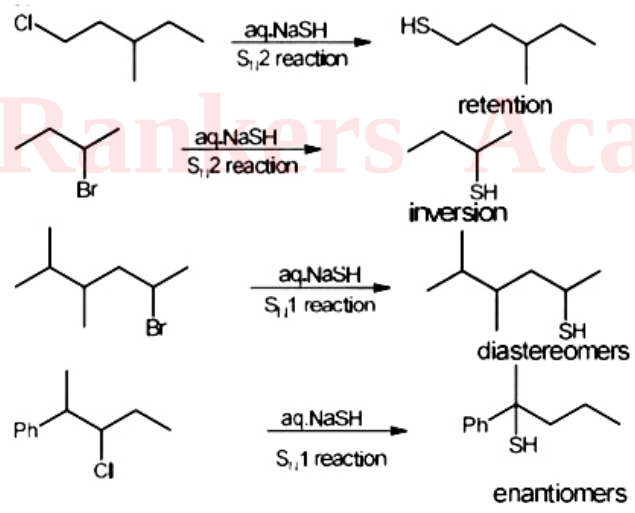
45. The cell reaction is

$MnO_4^{2-}+5Fe^{2+}+8H^+\rightarrow Mn^{2+}+5Fe^{3+}+4H_2O$

$E_{Cell}^0=1.51-0.78=0.73\,V$

$E_{Cell}=0.73+\frac{0.06}{5}\log[H^+]^8=0.73-0.48=0.25V$

46.



47. Coceptual

48. $P_2O_3+3H_2O\rightarrow2H_3PO_3$

$P_4+3NaOH+3H_2O\rightarrow PH_3+3NaH_2PO_2$

$PCl_5+CH_3COOH\rightarrow CH_3COCl+POCl_3+HCl$

$H_3PO_2+2H_2O+4AgNO_3\rightarrow H_3PO_4+4Ag+4HNO_3$

HCl can also act as reducing agent

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